Understanding the temporal effects of growth genes in wheat

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Background

- Growth stage timing is an essential component of regional adaptation.
- Photoperiod and vernalization response are major controllers of growth stage transitions but do not explain all variation.
- Relative height of plants across the growing season is associated with growth stage timing and with regional adaptation.
- UAVs can rapidly and reproducibly phenotype whole fields at many timepoints, making season-wide studies more feasible.
- Existing planting design (yield plots, headrows) are not well suited to planting large mapping populations for UAV studies.

Germplasm

- Two biparental recombinant inbred line (RIL) populations.
- Hilliard by GA06493-13LE6 (UX1992): 191 RIL evaluated.
- Hilliard by TX12D4896 (UX2012): 168 RIL evaluated.
- Parents were assessed for major known growth gene alleles using KASP assays.
- Sequenced all lines with GBS and created biparental linkage maps.

Table 1: KASP assay results of parents for major growth genes. Hilliard is common between biparentals. Columns contain gene name, chromosome and Mb position of gene in CS refseq v2.1 in format chr_Mb.

Line name	*Ppd-A1* 2A_42	*Ppd-B1* 2B_63	*Ppd-D1* 2D_36	*Vrn-A1* 5A_589	*Vrn-B1* 5B_577	*Rht-D1* 4D_19	*Rht-25* 6A_148	*Rht-24* 6A_416
GA06493-13LE6	A1b[a]	B1b[a]	D1a[b]	Strong[c]	Strong[c]	D1b[e]	D[e]	B[e]
HILLIARD	A1b	B1b	D1a	Strong	Strong	D1b	a[f]	a[f]
TX12D4896	A1b	B1b	D1a	Weak[d]	Strong	D1b	d	b

a: Photoperiod sensitive allele
b: Photoperiod insensitive allele
c: Strong winter type
d: Weak winter type
e. Dwarfing allele
f. Wild type allele

Phenotyping

- Planted lines in a checkerboard: double headrows alternate with empty rows.
- Grown in Raleigh, NC and Kinston, NC in 2023 and 2024 (four environments).
- UAV equipped with RGB camera flown 6-11 times in each environment.
- Texas A&M extracted canopy cover (CC), excess greening (EG), canopy height (CH), canopy volume (CV) per flight.
- After each flight, manually collected height, aboveground dry biomass, and Zadok's growth stage for 4 replicates of each parent.

Figure 1: UAV experimental plot layout. Each set of paired plots is one genotype. Experiment is planted in the middle 10 columns (blue box). Replicates of each parent are planted in second and thirteenth columns; collected height, Zadok's growth stage measurements, and dry biomass from a random subset after each flight. First and fourteenth columns are replicates of each parent.

Data processing

- Cleaned raw UAV trait data, then fit splines for each plot.
- Used average Zadok's growth stage data from parents to assign a mean day for 8 growth stages (0, 32, 45, 55, 65, 75, 85) in each environment.
- Calculated BLUEs for each individual at each growth stage using a mixed linear model with an autocorrelated row:column residual within environment:

$$y = Xb_{Genotype} + Zu_{Environment} + \varepsilon_{row:column:Environment}$$

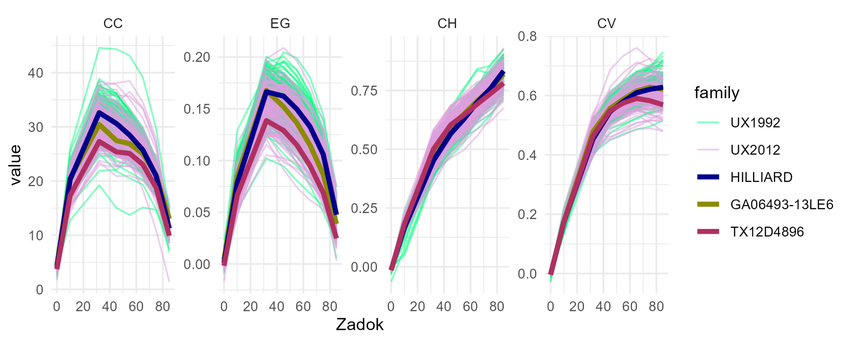

Figure 2: Plots of BLUEs for UAV data throughout the growing season for each of four traits. Plots are separated by trait: canopy cover (CC), excess greening (EG), canopy height (CH), and canopy volume (CV). Colors represent biparental families with thin lines and parents with thick. X-axis is Zadok's growth stage, y-axis is response.

QTL ID

- Performed linkage analysis on each biparental using r/qtl2.
- Used the BLUEs of each trait as phenotype.
- Each Zadok's growth stage timepoint considered independently.

Results

1. At *Vrn-A1* on 5A at 589 Mb, the TX12D4896 allele (weak winter type) confers early season increase in canopy volume, whole season increase in canopy height, and late season decrease in excess greening, indicating early senescence.
2. There is a QTL in UX1992 that is co-located or linked with *Vrn-A1*. The GA parent confers early season increase in canopy volume, early season increase in canopy height, and slight increase in early season excess greening but slight decrease in late season excess greening, indicating later senescence.
3. For reduced height gene *Rht-25* on 6A at 148 Mb, dwarfing allele confers whole-season decrease in canopy height, with a small late season decrease in canopy volume.
4. For height gene *Rht-24* on 6A at 416 Mb, the dwarfing allele confers a significant early season reduction in canopy height.

Many novel QTL detected. Notably:

- 1DS has an early effect QTL on canopy height at 56 Mb, and an early effect on canopy cover and a mid season effect on excess greening at 92 Mb in UX2012.
- 2BL has a QTL affecting early season canopy cover at 572 Mb, and one at 783 Mb affecting early season volume and height.
- A 3BL QTL spanning 804-848 Mb affects whole season canopy height and early season excess greening.
- A 7AL QTL at 677 Mb has an early and mid season effect on canopy cover.

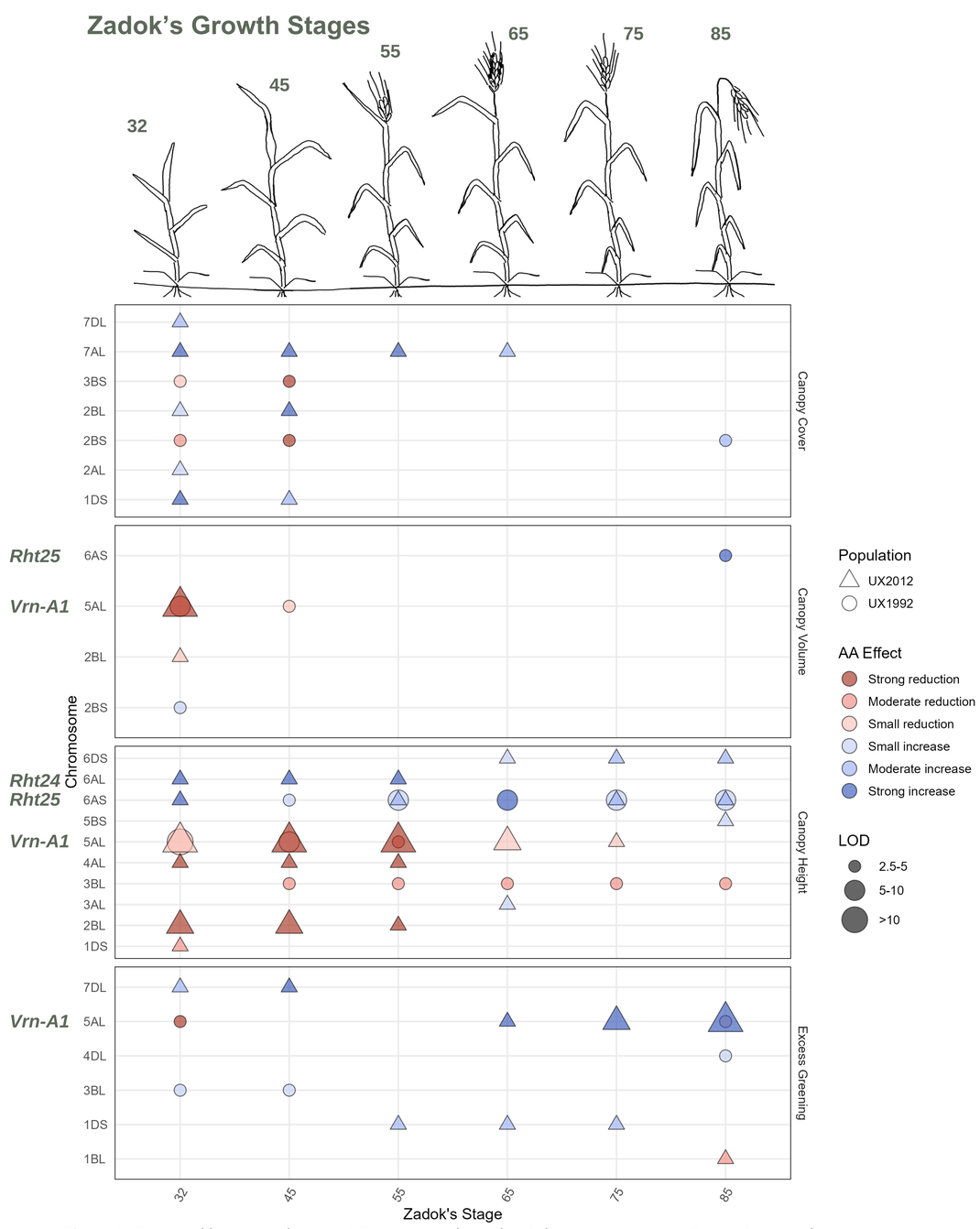

Figure 3: Temporal heatmap showing QTL presence throughout the growing season. Top pictogram shows representation of wheat at each growth stage. Zakok's 32 is between tillering and stem extension, 45 is boot just visibly swollen, 55 is inflorescence half emerged, 65 is anthesis halfway progressed, 75 is medium milk, and 85 is soft dough. Heatmaps shows significant QTL for each trait. Y-axis is chromosome arm, x-axis is Zadok's growth stage. Color and intensity of each shape indicates direction and magnitude of effect of the Hilliard allele within a trait, size indicates significance, shape indicates which biparental detected that QTL.

Conclusions

The major driver of differences in canopy volume and height is *Vrn-A1* in UX2012, and a novel QTL at or near the *Vrn-A1* locus in UX1992. Both QTL have similar effects, though the TX12D4896 allele has a stronger effect than the GA06493-13LE6 allele. Interestingly, the GA allele confers later senescence relative to the Hilliard allele, unlike the TX *Vrn-A1* allele.

In both populations, *Rht25* and *Rht24* significantly effect height through multiple growth stages as expected.

We detected major effect QTL for early and midseason growth on 2BL across several traits; Hilliard had the lower growth allele.

We observed novel QTL, notably: 1DS and 7AL with early and midseason effects on growth, and 3BL with a whole-season effect on canopy height.

We found many additional QTL with low significance and that only affect one trait at one or two timepoints.

Small Grains Genotyping

USDA

Wheat CAP Coordinated Agricultural Project

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